Taeyoung Kim

Ph.D. Candidate, School of Computing, KAIST

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Research Interests

Machine learning for mathematics, especially combinatorics: deep-learning frameworks for variational optimisation problems arising in extremal graph theory and in large deviations of random graphs, and more broadly for characterising extremisers of functional inequalities in harmonic analysis and PDEs. Earlier work on the theory of deep learning: infinite-width limits, in-context learning, and feature learning.

Education

Ph.D., School of Computing, KAIST — Aug 2023 – Feb 2027 (expected)
Advisor: Prof. Hongseok Yang
Dissertation (tentative): *Implicit Neural Representations for Variational Problems on Graphons*

M.S., School of Computing, KAIST — Sep 2021 – Aug 2023
Advisor: Prof. Hongseok Yang
Thesis: *An Infinite-Width Analysis on the Jacobian-Regularised Training of a Neural Network*

B.S., School of Computing and Department of Mathematical Sciences, KAIST
Double major in Computer Science and Mathematics

Publications

Conference Papers

[C2] **Taeyoung Kim**, Jineon Baek, Joonkyung Lee, Hongseok Yang. **Implicit Neural Representations for Variational Problems on Graphons.** *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. To appear. (25.7%, 7900/30709) [arXiv]

[C1] **Taeyoung Kim**, Hongseok Yang. **An Infinite-Width Analysis on the Jacobian-Regularised Training of a Neural Network.** *International Conference on Machine Learning (ICML)*, PMLR 235:24584–24657, 2024. (27.5%, 2609/9473) [PMLR] [arXiv]

Workshop Papers

[W2] **Taeyoung Kim**, Hongseok Yang. **Transformers Can Perform Distributionally-robust Optimisation through In-context Learning.** *ICML 2024 Workshop on In-Context Learning*, 2024. [OpenReview]

[W1] Jaehui Hwang, **Taeyoung Kim**, Hongseok Yang. **Analysing Feature Learning of Gradient Descent Using Periodic Functions.** *ICML 2024 Workshop on High-dimensional Learning Dynamics (HiLD)*, 2024. [OpenReview]

Preprints

[P2] Seonghyuk Im, **Taeyoung Kim**, Hongseok Yang. **Goodman-type Lower Bounds for Odd Cycles I: The Dense Regime.** arXiv:2610.18277, 2026. [arXiv]

[P1] Sangho Lim, Seonghyuk Im, **Taeyoung Kim**, Kyeongsik Nam, Hongseok Yang. **Bipodal Optimizers in the Upper-Tail Variational Problem for Regular Subgraph Densities.** arXiv:2609.15222, 2026. [arXiv]

Talks

Implicit Neural Representation of Graphons for Extremal Graph Theory — Jul 2026
Tutorial, *PACOH Workshop: Artificial Intelligence for Combinatorics*, Jeju, Korea

Transformers Can Perform Distributionally-robust Optimisation through In-context Learning Aug 2024
Seminar, *Computational Statistics and Machine Learning Group*, Department of Statistics, University of Oxford, UK

Honors and Awards

Top Reviewer (top 10% of reviewers), NeurIPS 2026 2026

Gold Reviewer (top 25% of reviewers), ICML 2026 2026

Outstanding TA Award, KAIST Spring 2022
For *Introduction to Algorithms*

Notes

[N4] **Taeyoung Kim**. **Goodman-type Lower Bounds for Graph Homomorphism Densities.** Note with a Lean 4 formalisation, 2026. [PDF] [Lean]

[N3] **Taeyoung Kim**. **Commonness of Independent Apices of Trees and Even Cycles.** Note with a Lean 4 formalisation, 2026. [PDF] [Lean]

[N2] **Taeyoung Kim**. **Semi-inducibility of Alternating Cycles with Length $4k+2$.** Note with a Lean 4 formalisation, 2026. [PDF] [Lean]

[N1] **Taeyoung Kim**. **Off-diagonal Commonality of a Shorter Even Cycle and a Longer Odd Cycle.** Note with a Lean 4 formalisation, 2026. [PDF] [Lean]

Teaching

Teaching Assistant, KAIST

Introduction to Logic for Computer Science Spring 2023, Spring 2024, Spring 2025

Operating Systems and Lab Fall 2024

Special Topics in Computer Science: Computational Learning Theory Fall 2023

Introduction to Algorithms Fall 2021, Spring 2022

Professional Service

Reviewer

- International Conference on Learning Representations (ICLR), 2026, 2027
- International Conference on Machine Learning (ICML), 2026
- Conference on Neural Information Processing Systems (NeurIPS), 2026
- Transactions on Machine Learning Research (TMLR), 2026–present

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